

ORIGINAL ARTICLE

Effects of dietary supplementation with herbal extract mixture on growth performance, organ weight and intestinal morphology in weaning piglets

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Abstract

Background: Many countries are increasingly prohibiting the addition of antibiotics in livestock diets. Therefore, herb extracts have gradually drawn attention to substitute antibiotics. Our present study aimed to determine the effects of herbal extract mixture (HEM) in dietary on growth performance, organ weight, intestinal morphology and intestinal nutrient transporters in weaned pigs.

Methods: 27 piglets (Duroc × [Landrace × Yorkshire]; Body Weight (BW) = 5.99 ± 0.13 kg) were weaned at day 21 and randomly divided into three groups (n = 9 piglets/group). All piglets received a basal diet containing similar amounts of nutrients for 14 days. The three groups were the control (no additive), the antibiotics (375 mg/kg chlortetracycline, 20%, 500 mg/kg enramycin, 4%, 1,500 mg/kg oxytetracycline calcium, 50%) and the HEM group (1000 mg/kg extract mixture of golden-and-silver honeysuckle, huangqi, duzhong leaves and dangshen). After 14 d of treatment, we collected tissue samples to measure organ weight, intestinal parameters, intestinal morphology, digestive enzyme activities and intestinal mRNA expression of nutrient transporters.

Results: The HEM group had no effects on growth performance and organ weight of weaned pigs. But compared with the control group, both HEM and antibiotics improved intestinal morphology, and HEM elevated the expression of nutrient transporters in ileum (SLC6A9, SLC15A1, and SLC5A1). HEM significantly decreased the activities of maltase in ileum and the ratio of small intestinal weight to BW than control group.

Conclusions: These results indicate benefit effects of the supplementation of HEM in diet, including modulating intestinal morphology and increasing the mRNA expression of nutrients transporters. These findings suggest that HEM provides novel insights into a variety of herbal extract mixtures to replace antibiotics in animal production.

KEYWORDS

growth performance, herbal extract mixture, intestinal morphology, weaning piglets

1 | INTRODUCTION

According to many countries' instruction, antibiotics have been reduced or prohibited within the livestock diet. Thus, it is a pivotal measure to search for a valid solution that can replace antibiotics in animal production. Recently, many studies have shown that herbs or herbal extracts have multiple characteristics with anti-bacterial, anti-inflammatory, and anti-viral activities (Kong et al., 2007b; Si et al., 2006). Golden-and-silver honeysuckle (*Lonicera japonica* Thunb), huangqi (*Astragalus membranaceus*), duzhong leaves (*Eucommia folium*) and dangshen (*Codonopsis pilosula*) have been applied to Chinese medicine cure for a long time. Increasing evidence has showed that Chinese herbal extract plays immunomodulatory and antioxidative roles as medicines or supplements in human health and food industry (Borchers, Hackman, Keen, Stern, & Gershwin, 1997; Zhu et al., 2004). Many studies have reported that HEM has a positive effect in livestock production, such as broilers, pigs and calves (Bombik, Bombik, Frankowska, Trawinska, & Saba, 2012; Bozkurt et al., 2012; Yan, Meng, & Kim, 2012). For example, feeding herbal extract in diet can improve the growth performance or reduce the proliferation of coliform bacteria in the gut of weaning piglets (Li et al., 2008; Namkung, Li J. Gong, Yu, Cottrill, & de Lange, 2004). In addition, our previous study has found that dietary Chinese herbal powder affects growth performance, serum metabolic products and intestinal health status in early-weaning pigs (Kong et al., 2007a). The constituents associated with the activity component of *stragalus membranaceus* are isoflavonoids, saponins, polysaccharides and various trace elements (Sinclair, 1998; Wagner et al., 1997). Previous studies have reported that extracts of Duzhong leaves have a protective effect on biomolecules, which are damaged by oxidative stress (Xiao et al., 2019; Yuan et al., 2017). *Codonopsis pilosula* exhibits tumour-resistant, diabetic-resistant, antiaging and inflammation-resistant properties (He et al., 2015, 2016). A variety of herbal extracts can have accumulative or synergistic effects on animal health. In the livestock production, Chinese herbal extracts are regarded as safe supplements to prevent weaning-associated intestinal dysfunction and improve growth performance in piglets (Kong et al., 2007a; Kong et al., 2007b).

Until now, the combined influence of herbal extract mixture on intestinal growth parameters, intestinal morphology and intestinal absorption have not been fully studied in weaned pigs. Hence, our

current study aimed to test the assumption that dietary addition of herbal extract mixture (HEM, golden-and-silver honeysuckle, huangqi, duzhong leaves, dangshen) could improve intestinal absorption in weaning piglets. In addition, we also compared the effects of HEM and antibiotics on intestinal absorption.

2 | MATERIALS AND METHODS

The whole experimental procedures in the current study were supervised and allowed by the Animal Care and Use Committee of Hunan Normal University, Changsha City, Hunan, China (2019-1A). The study was conducted in the Animal Research Facility at Hunan Normal University, China.

2.1 | Experimental design and treatments

Twenty-seven piglets (Duroc × [Landrace × Yorkshire]; Body Weight (BW) = 5.99 ± 0.13 kg) were weaned at day 21 and randomly divided into three treatments ($n = 9$ piglets per group): (a) control treatment (con): pigs were raised a basal diet; (b) antibiotics treatment: pigs were raised a basal diet supplemented with 375 mg/kg chlortetracycline (20%), 500 mg/kg enramycin (4%) and 1,500 mg/kg oxytetracycline calcium (50%); and (c) herbal extract mixture (HEM) treatment: pigs were raised a basal diet supplemented with 1,000 mg/kg HEM. The dosage for HEM supplementation was selected referring to previous studies (Kong et al., 2007a). The basal diet attained the NRC (2012) nutrient standard for piglets during the whole experimental period, and the diet composition was shown in the Table 1. The herbal extract was a mixture from golden-and-silver honeysuckle (*Lonicera japonica* Thunb), huangqi (*Astragalus membranaceus*), duzhong leaves (*Eucommia folium*) and dangshen (*Codonopsis pilosula*). HEM was supplied by Anhui Tianan Biotechnology Company Limited, in Luan city, China. There were nine pens per treatment and one piglet per pen. The experimental environment was performed as described (Chen et al., 2019). All piglets had ad libitum access to dietary and drinking water at whole experimental time. Growth performance of average daily gain (ADG), average daily feed intake (ADFI) and food: gain (F: G) were recorded (Yang, Xiong, Li & Yin, 2016).

TABLE 1 Composition of diets (as-fed basis, %)

Item	Dietary treatment ^a , %		
	CON	Antibiotics	HEM
Ingredient			
Corn	28.33	28.1	28.23
Extruded corn	30	30	30
Soy protein concentrate	6	6	6
Soya bean meal, 43% CP	8	8	8
Fish meal, 63% CP	5	5	5
Whey	15	15	15
Glucose	3	3	3
Soya bean oil	0.5	0.5	0.5
Limestone	0.86	0.86	0.86
Calcium hydrogen phosphate	0.4	0.4	0.4
Choline chloride, 50%	0.1	0.1	0.1
Antioxidants ^b	0.05	0.05	0.05
Citric acid	0.8	0.8	0.8
Salt	0.1	0.1	0.1
Vitamin-mineral premix ^c	0.45	0.45	0.45
Herbal extract mixture	0	0	0.10
Chlortetracycline, 20%	0	0.0375	0
Enramycin, 4%	0	0.05	0
Oxytetracycline calcium, 50%	0	0.15	0
Zinc oxide	0.2	0.2	0.2
L-Lysine, 98%	0.58	0.58	0.58
D-L-Methionine	0.38	0.38	0.38
L-Threonine	0.19	0.19	0.19
L-Tryptophan	0.06	0.06	0.06
Total	100.00	100.00	100.00
Calculated composition			
CP	18	18	18
ME, MJ/kg	14.2	14.2	14.2
Lysine ^d	1.35	1.35	1.35
Methionine ^d	0.39	0.39	0.39
Methionine + Cysteine ^d	0.74	0.74	0.74
Threonine ^d	0.79	0.79	0.79
Tryptophan ^d	0.22	0.22	0.22

^aCON, basic diet; HEM, basic diet supplemented with 1,000 mg/kg herbal extract mixture; Antibiotics, diet supplemented with 375 mg/kg Chlortetracycline (20%), 500 mg/kg Enramycin (4%) and 1,500 mg/kg Oxytetracycline calcium (50%).

^bAntioxidants contained 60g/kg butylhydroquinone and 180g/kg ethoxyquinoline.

^cVitamin-mineral premix supplied per kilogram of feed: 10,000 IU of vitamin A, 1,000 IU of vitamin D₃, 80 IU of vitamin E, 2.0 mg of vitamin K₃, 0.03 mg of vitamin B₁₂, 12 mg of riboflavin, 40 mg of niacin, 25 mg of d-pantothenic acid, 0.25 mg of biotin, 1.6 mg of folic acid, 3.0 mg of thiamine, 2.25 mg of pyridoxine, 300 mg of choline chloride, 150 mg of Fe (FeSO₄), 100 mg of Zn (ZnSO₄), 30 mg of Mn (MnSO₄), 25 mg of Cu (CuSO₄), 0.5 mg of I (KIO₃), 0.3 mg of Co (CoSO₄), 0.3 mg of Se (Na₂SeO₃) and 4.0 mg of ethoxyquin.

^dStandardized ileal-digestible.

2.2 | Sample collection

On days 14 of experiment, the BW of piglets was determined after fasting overnight, collecting the blood and tissue sample from 9 piglets of each group according to the procedure (Yang, Xiong, et al., 2016). Piglets were disposed in the experiment as described (Zong et al., 2018). Then, the mesenterium peeled off, the small intestine (SI) and large intestine (LI) were taken out and their length measured used a ruler by fixing it vertically in the experimental table. The weight of SI and LI was determined after removal of content and mesenterium. Absolute weight and absolute length of intestine were defined as the weight and length of intestine by direct measured. Relative weight and relative length of intestine were defined as the weight and length of intestine by calculated in final BW. The weight of organs (heart, liver, kidneys, stomach and spleen) was measured. The pH value of gastrointestinal content (stomach, ileum, caecum and colon) was detected by the handheld pH meter. The intestinal sample were collected as described in the part of materials and methods (Zong et al., 2018). In this study, the intestinal sample were fixed in 4% paraformaldehyde reagent to assess the intestinal morphology, and intestinal mucosa was treated to analyse mRNA expression of nutrients carriers and digestive enzyme activity.

2.3 | Measurement of intestinal morphology

Intestinal histology was performed as previous study did (Zong et al., 2018). Briefly, intestinal tissue was dehydrated in a gradual concentration of ethanol and embedded in paraffin wax. Each embedded tissue was cut differently three sections (5 µm in thickness) and then placed on the microslides. The microslides were stained with haematoxylin and eosin. Villus height (considered as the peak of the villi to the villus crypt joint) and crypt depth (considered as the depth of emboly between close villi) of each sample were acquired and measured at 10 × magnification by optical microscope (Leica Imaging Systems Ltd.). Each piglet of villus height and crypt depth was calculated from at least 30 intact villi and their associated crypts.

2.4 | Analysis of digestive enzyme activities

The jejunum and ileum were grinded by liquid nitrogen and were extracted in ice-cold phosphate-buffered saline, and then, it was centrifuged for 10 min (3,000 g, 4°C). The intestinal digestive enzymes including alkaline phosphatase (ALP), lactase, sucrase and maltase were determined by commercially available kits referring to the company' instructions (Nanjing Jiancheng Bioengineering Institute). The protein concentration of jejunum and ileum was analysed by a bicinchoninic acid protein assay kit (Beyotime Biotechnology).

2.5 | Gene expression analysis

RNA extract, cDNA synthesis, the procedure of qPCR and the calculated method to analyse the relative mRNA expression as described according to our previous study (Chen et al., 2019). Primers performed in present study were designed with NCBI according to the gene sequence of *Sus Scrofa* on GeneBank (Table S1). TRIZOL reagent and cDNA synthesis kit were purchased from the company of Takara. SYBR Green mix was purchased from the company of Thermo Fisher Scientific.

2.6 | Statistical analysis

Data in current study were showed as means $\pm$ SEM and analysed by one-way ANOVA using the 22th version of SPSS software (SPSS Inc.). Differences among treatment means were explored using the test of Duncan's range. Tables in the present study were produced by Word 2016 software (Microsoft Inc), and the Figure 1 was generated by Graphpad Prism 6.0 (GraphPad Software, Inc.). *P*-values <0.05 were indicated statistical significance in this study, and $0.05 \leq P$ -values <0.10 were termed as the trend for statistically significant.

3 | RESULTS

3.1 | Growth performance and organ weight

We determined the influence of HEM on growth performance in weaning piglets. Among the three dietary treatments, no significant

differences were found in Table 2. Next, we explored the weight of the organs, including liver, spleen, kidneys, heart and stomach. Compared with the HEM group and control group, antibiotics group tended to elevate the weight of the liver ($p = .076$) and kidneys ($p = .054$). However, no significant differences were detected between the HEM treatment and the control group (Table 3).

3.2 | Intestinal growth parameters and gastrointestinal content pH

To examine the influences of feed HEM addition on intestinal growth characters and gastrointestinal content pH value, we measured the intestinal growth parameters and the pH value of digestive tract content (Table 4, Table S2). In comparison with the control treatment, feed HEM addition markedly decreased the relative small intestinal weight ($p = .50$). The HEM group and the antibiotics group were similar in the absolute large intestinal length and the relative small intestinal weight. Nevertheless, no significant effects were implied in the absolute small intestinal weight, absolute small intestinal length, relative large intestinal weight and relative large intestinal length among the three dietary treatments. Moreover, compared with control group, dietary HEM supplementation had no effect on the pH value of ileum, colon, caecum and stomach.

3.3 | HEM on intestinal morphology

In order to further explore the influences of dietary HEM supplementation on intestinal development, we examined the morphology

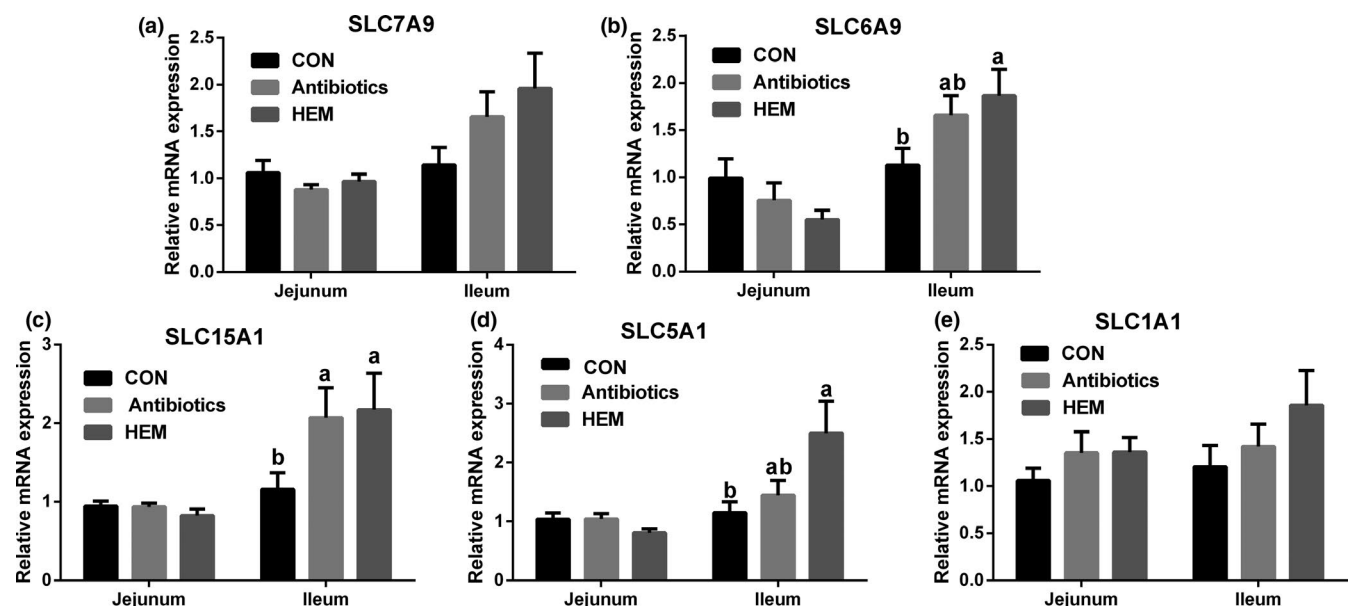


FIGURE 1 Effects of dietary supplementation with HEM on the expression of nutrient transporter genes in weaning piglets. CON, basic diet; HEM, basic diet supplemented with 1,000 mg/kg herbal extract mixture; Antibiotics, diet supplemented with 375 mg/kg Chlortetracycline (20%), 500 mg/kg Enramycin (4%) and 1,500 mg/kg Oxytetracycline calcium (50%). The expression of SLC7A9 (a), SLC6A9 (b), SLC15A1 (c), SLC5A1 (d) and SLC1A1 (e) was measured by qRT-PCR. Values are expressed as mean $\pm$ SEM, $n = 10$. ^{a,b}Means of the bars with different letters are significantly different among groups ($p < .05$)

Item	Dietary treatment ^b			P-value
	CON	Antibiotics	HEM	
BW, kg				
Initial	6.00 ± 0.28	5.98 ± 0.32	5.98 ± 0.26	1.000
Final	8.12 ± 0.45	8.14 ± 0.48	7.86 ± 0.42	.950
ADFI, g				
d 0 to 7	185.35 ± 13.60	208.28 ± 20.38	189.43 ± 19.20	.440
d 8 to 14	501.91 ± 36.37	539.04 ± 55.02	476.19 ± 41.87	.746
d 0 to 14	331.46 ± 23.19	360.94 ± 36.22	321.92 ± 27.94	.654
ADG, kg/d				
d 0 to 7	0.07 ± 0.01	0.05 ± 0.03	0.06 ± 0.02	.460
d 8 to 14	0.27 ± 0.02	0.31 ± 0.03	0.25 ± 0.03	.500
d 0 to 14	0.18 ± 0.02	0.18 ± 0.03	0.16 ± 0.02	.820
F: G, g/g				
d 0 to 7	2.51 ± 0.43	3.20 ± 1.09	3.29 ± 0.60	.716
d 8 to 14	1.85 ± 0.06	1.77 ± 0.13	1.87 ± 0.08	.663
d 0 to 14	1.93 ± 0.10	1.97 ± 0.17	2.22 ± 0.20	.574

^aValues are expressed as mean ± SEM; *n* = 9.

^bCON, basic diet, HEM, basic diet supplemented with 1,000 mg/kg herbal extract mixture; Antibiotics, diet supplemented with 375 mg/kg Chlortetracycline (20%), 500 mg/kg Enramycin (4%) and 1,500 mg/kg Oxytetracycline calcium (50%).

TABLE 2 Effects of dietary supplementation with herbal extract mixture (HEM) on growth performance of weaned piglets^a

Item	Dietary treatment ^b			P-value
	CON	Antibiotics	HEM	
Liver, g	207.60 ± 10.25	217.25 ± 18.75	192.14 ± 9.85	.579
Spleen, g	17.23 ± 1.71	17.23 ± 1.86	19.08 ± 1.14	.754
Kidneys, g	50.62 ± 2.85	56.74 ± 5.24	52.18 ± 4.63	.478
Heart, g	44.88 ± 2.71	46.69 ± 3.60	47.32 ± 2.85	.894
Stomach weight, g	53.64 ± 2.30	55.89 ± 2.74	53.83 ± 3.65	.935
Liver/BW, g/kg	25.66 ± 0.58	26.42 ± 1.13	23.34 ± 0.95	.076
Spleen/BW, g/kg	2.11 ± 0.16	2.11 ± 0.20	2.30 ± 0.07	.440
Kidneys/BW, g/kg	6.24 ± 0.18	6.57 ± 0.24	6.20 ± 0.17	.054
Heart/BW, g/kg	5.52 ± 0.14	5.69 ± 0.17	5.71 ± 0.17	.89
Stomach weight/ BW, g/kg	6.68 ± 0.27	6.94 ± 0.30	6.47 ± 0.15	.523

^aValues are expressed as mean ± SEM; *n* = 9.

^bCON, basic diet, HEM, basic diet supplemented with 1,000 mg/kg herbal extract mixture; Antibiotics, diet supplemented with 375 mg/kg Chlortetracycline (20%), 500 mg/kg Enramycin (4%) and 1,500 mg/kg Oxytetracycline calcium (50%).

TABLE 3 Effects of dietary supplementation with herbal extract mixture (HEM) on organ weight of weaned piglets^a

of small intestine and large intestine by HE staining (Table 5, Table S3). These data indicated that both the HEM group and the antibiotics group significantly decreased the duodenal crypt depth (*p* = .005) and significantly increased the ratio of villus height to crypt depth for the duodenum (*p* = .008) and jejunum (*p* = .036) versus the control group. But there is similar influence in crypt depth of colon and caecum among the three dietary treatments.

3.4 | Intestinal enzyme activities

The results of intestinal enzyme activities were showed in Table 6. HEM significantly decreased the enzyme activity of ileal maltase (*p* = .001) relative to control group, but had no effect on the enzyme activities of ALP, lactase and sucrase for jejunum and ileum among three groups.

TABLE 4 Effects of dietary supplementation with herbal extract mixture (HEM) on intestinal growth parameters of weaned piglets¹

Item	Dietary treatment ²			P-value
	CON	Antibiotics	HEM	
SI length, cm	936.66 ± 20.77	1,005.62 ± 32.77	937.22 ± 23.89	.239
SI weight, g	403.03 ± 20.04	383.60 ± 22.72	378.36 ± 28.28	.508
LI length, cm	167.00 ± 4.96	181.87 ± 4.99	182.77 ± 6.57	.138
LI weight, g	140.00 ± 6.67	154.27 ± 9.92	145.07 ± 8.66	.715
SI length/BW, cm/kg	117.24 ± 4.77	125.37 ± 5.06	115.08 ± 5.15	.515
SI weight/BW, g/kg	49.92 ± 1.74 ^a	47.13 ± 1.20 ^{ab}	45.35 ± 1.26 ^b	.050
LI length/BW, cm/kg	20.92 ± 1.02	21.47 ± 0.55	22.59 ± 1.51	.698
LI weight/BW, g/kg	17.36 ± 0.56	19.04 ± 0.78	17.54 ± 0.65	.264

Values in the same row not sharing a common superscript (a,b) differ significantly.

Abbreviations: LI = Large intestine; SI = Small intestine.

¹Values are expressed as mean ± SEM; *n* = 9.

²CON, basic diet; HEM, basic diet supplemented with 1,000 mg/kg herbal extract mixture; Antibiotics, diet supplemented with 375 mg/kg Chlortetracycline (20%), 500 mg/kg Enramycin (4%) and 1,500 mg/kg Oxytetracycline calcium (50%).

TABLE 5 Effects of dietary supplementation with herbal extract mixture (HEM) on the intestinal morphology of weaned piglets¹

	Dietary treatment ²			
Item	CON	Antibiotics	HEM	P-value
Villus Height, μm				
Duodenum	246.42 ± 22.13	289.92 ± 12.62	256.76 ± 11.98	.114
Jejunum	362.07 ± 12.25	377.43 ± 21.42	383.09 ± 13.83	.597
Ileum	277.73 ± 9.16	289.99 ± 11.17	304.17 ± 13.19	.434
Crypt Depth, μm				
Duodenum	376.19 ± 4.29 ^a	300.08 ± 19.47 ^b	289.65 ± 11.82 ^b	.005
Jejunum	228.65 ± 7.48	217.40 ± 4.56	208.38 ± 8.95	.166
Ileum	221.66 ± 9.66	224.68 ± 9.50	231.70 ± 7.51	.725
Ratio of villus height to crypt depth				
Duodenum	0.68 ± 0.06 ^b	1.02 ± 0.04 ^a	0.94 ± 0.08 ^a	.008
Jejunum	1.66 ± 0.06 ^b	1.91 ± 0.01 ^a	1.94 ± 0.08 ^a	.036
Ileum	1.37 ± 0.08	1.40 ± 0.08	1.40 ± 0.06	.956

Values in the same row not sharing a common superscript (a,b) differ significantly.

¹Values are expressed as mean ± SEM; *n* = 9.

²CON, basic diet; HEM, basic diet supplemented with 1,000 mg/kg herbal extract mixture; Antibiotics, diet supplemented with 375 mg/kg Chlortetracycline (20%), 500 mg/kg Enramycin (4%) and 1,500 mg/kg Oxytetracycline calcium (50%).

3.5 | Intestinal nutrient transporter genes

To investigate the influences of dietary HEM on intestinal nutrients transporters, we performed qPCR analysis to measure the gene expression of intestinal nutrients transporters (SLC7A9, SLC6A9, SLC15A1, SLC5A1 and SLC1A1) in the jejunum and ileum. As described in Figure 1, in comparison with control group, the HEM treatment markedly elevated ileal mRNA abundance of SLC6A9, SLC15A1

and SLC5A1. However, no different effects of jejunal nutrients transporters were presented among three dietary types.

4 | DISCUSSION

HEM has been widely demonstrated to involve growth performance in the piglets (Hanczakowska, Świątkiewicz, & Grela, 2015;

	Dietary treatment ²			
Item	CON	Antibiotics	HEM	P-value
Alkaline phosphatase, U/mg protein				
Jejunum	136.27 ± 24.93	205.97 ± 35.98	164.98 ± 12.13	.437
Ileum	281.79 ± 55.00	286.31 ± 56.20	371.45 ± 55.36	.444
Maltase, U/ mg protein				
Jejunum	173.21 ± 24.73	183.58 ± 27.31	174.11 ± 24.86	.953
Ileum	177.14 ± 35.25 ^a	75.18 ± 10.64 ^b	55.07 ± 10.67 ^b	.001
Lactase, U/mg protein				
Jejunum	21.39 ± 4.05	24.20 ± 5.05	21.03 ± 3.53	.151
Ileum	7.31 ± 1.59	3.41 ± 1.66	4.13 ± 0.90	.151
Sucrase, U/mg protein				
Jejunum	42.14 ± 7.52	41.76 ± 12.29	26.45 ± 6.95	.372
Ileum	27.66 ± 8.56	34.17 ± 6.91	28.96 ± 6.77	.817

Values in the same row not sharing a common superscript (a,b) differ significantly.

¹Values are expressed as mean ± SEM; n = 9.

²CON, basic diet, HEM, basic diet supplemented with 1,000 mg/kg herbal extract mixture; Antibiotics, diet supplemented with 375 mg/kg Chlortetracycline (20%), 500 mg/kg Enramycin (4%), and 1,500 mg/kg Oxytetracycline calcium (50%).

TABLE 6 Effects of dietary supplementation with herbal extract mixture (HEM) on digestive enzyme activities of weaned piglets¹

Namkung et al., 2004; Yan, Meng, & Kim, 2011). Indeed, Yan et al., (2012) reported that dietary HEM supplementation had a higher ADG in 4–6 weeks of weaning pigs and growing pigs. Meanwhile, Yan et al., (2012) investigated that dietary HEM supplementation had no significant difference on the growth performance during 2–3 weeks of weaning piglets. In this study, dietary HEM had similar growth performance of weaning piglets during the experimental treatments. These differences in growth performance may partly be due to the experimental days tested. In our present study, we explored that HEM did not affect to growth performance in 2 weeks after weaning, as same as the Yan et al., (2012) research. Young animals had a serious injury in intestinal development and digestive activities of intestine after weaning (Ren et al., 2018). The organ weight of pig was estimated to be in good health, disease free and a resting state (Upton, 2008). Diet supplementation of Korean aged garlic extract had decreased the liver weight compared with the negative control diet. Other studies reported that the relative organ weight of broilers such as liver, spleen, were not affected by the supplementation of herbal extract in the diet (Begum, Hossain, & Kim, 2014; Cabuk, Bozkurt, Alcicek, Akba___, & Küçükyılmaz, 2006). Our data in this study reported that antibiotics supplementation tended to increase the weight of liver and kidneys, but the control dietary and the HEM dietary had a similar influence to organ weight, suggesting that 1000 mg/kg HEM supplementation in diet is a safe dose to organ development.

It is well known that dietary regulate intestinal mucosal growth (Dowling, 1982). Özek, Wellmann, Ertekin, & Tarim, (2011) reported that dietary herbal essential oil mixture increased small intestine weight in laying hens. Hernandez, Madrid, Garcia, & Orengo, (2004) reported that dietary two plant extracts supplementation had no significant difference of large or small intestinal weight in broilers.

Our results showed that dietary HEM supplementation decreased relative small intestine weight. This might be because different plant extract have distinctive effect on intestinal growth parameters in livestock, such as laying hens, broilers and swine.

The intestinal morphology can provide some information on gastrointestinal development (Xia et al., 2019). Herbal essential oils in diet improved the villus width and height in the jejunum and ileum of quail breeders (Hajiaghapour & Rezaeipour, 2018). Broilers fed on the organic acid blends and essential oil blends had higher villus height and lower crypt depth in the ileum (Basmacioğlu-Malayoğlu, Ozdemir, & Bağrıyanık, 2016). Consistent with those earlier studies, we observed that HEM improved intestinal morphology through increased the ratio of villus height to crypt depth in two small intestine segments of duodenum and jejunum, and HEM had the same effect to intestinal morphology with antibiotics. It is indicating that HEM has the likely beneficial effect with antibiotics to intestinal morphology. In addition, Shams, Dastar, Zerehdaran, Khomeiri, & Moradi, (2012) reported that the crypt depth would be deeper when the weight of the small intestine would be heavier, which is consistent with our present result of HEM decreasing with small intestinal weight and crypt depth at same time.

The activities of most digestive enzymes are declined due to the challenge of weaning stress (See, Epstein, Dahl, & Bass, 1990). Wang, Yang, Wang, Yang, & Cui, (2011) showed that herbal extract increased post-ruminal digestive enzymes activity in beef cattle. The rate of development of the digestive enzymes might be affected by the dietary intake of their specific substrates (Jiang, Ko, Kang, & Lee, 2007). Our results showed that HEM decreased the activity of maltase in ileum and had no effect in other digestive enzymes. This might be because HEM could influence the substrate of maltase, resulting in reducing the activity of maltase in ileum.

The small intestine is the main location to obtain nutrients in pigs, such as fatty acid, glucose and peptides (Lindemann, Cornelius, El Kandelgy, Moser, & Pettigrew, 1986). In addition, the small intestine modulates the transport of amino acid and provides the requirement of amino acids to all tissue (Alexander & Carey, 1999; Ren et al., 2017). SLC15A1 presents abundant peptides to supply the least demand for amino acids (Bröer & Fairweather, 2019). SLC6A19 is the important neutral amino acid transporters located at the intestinal apical membrane (Bröer et al., 2004). SLC1A1 is NA^+ /glucose cotransporter across the intestinal brush border membrane (Wright, Loo, & Panayotova-Heiermann, 1998). In this study, HEM obviously elevated the mRNA level of SLC15A1, SLC6A9 and SLC1A1 in the ileum mucosa, suggesting that HEM improved the content of peptides, neutral amino acid and glucose of ileum to increase the absorptive capacity in weaning pigs.

5 | CONCLUSIONS

In summary, the HEM in feed had no effects on growth performance and organ weight. But HEM and antibiotics groups improved intestinal morphology and genes expression of nutrients transporters in weaning piglets. HEM significantly decreased relative small intestinal weight and the activities of maltase in ileum. These findings provide novel insights into HEM to replace antibiotics in animal production.

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CONFLICTS OF INTEREST

The authors have no conflicts of interest.

ANIMAL WELFARE STATEMENT

All animal experimental procedures in this study were reviewed and approved by the Animal Care and Use Committee of Hunan Normal University, Changsha City, Hunan, China (2019-1A). The study was conducted in the Animal Research Facility at Hunan Normal University, China.

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SUPPORTING INFORMATION

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